Anindya Mondal

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Education

Surrey Institute for People-centred AI, CVSSP, University of Surrey — **Guildford, United Kingdom**
PhD Candidate in Artificial Intelligence — *Oct. 2022 – present*
Research focus: Integrating auxiliary information for representation learning in natural world scenes.

Jadavpur University — **Kolkata, India**
B.E. (Hons.) in Electronics & Telecommunication Engineering, GPA: 8.79/10 — *Aug. 2018 – Jun. 2022*

Research Focus

2023–2024: Vision–language integration, multimodal learning, action recognition, text-to-image synthesis, object counting, 3D/4D generation.

2020–2022: Graph neural networks, time-series analysis, graph signal processing, neuromorphic vision, subspace learning.

Publications

Preprint

Anindya Mondal *et al.*, "CountLoop: Training-Free High-Instance Image Generation via Iterative Agent Guidance." (in preparation).

Published

AAAI '25: **Anindya Mondal**, Sauradip Nag, Xiatian Zhu, Anjan Dutta, "OmniCount: Multi-label Object Counting with Semantic–Geometric Priors." Proceedings of AAAI 2025. 10.1609/aaai.v39i18.34151

ICCVW '23: **Anindya Mondal**, Sauradip Nag, Joaquin M. Prada, Xiatian Zhu, Anjan Dutta, "Actor-Agnostic Multi-label Action Recognition with Multi-modal Query." ICCV Workshops 2023. 10.1109/ICCVW60793.2023.00086

ICASSP '23: J. A. C. Correa, J. H. Giraldo, **Anindya Mondal** *et al.*, "Time-Varying Signals Recovery via Graph Neural Networks." ICASSP 2023. 10.1109/ICASSP49357.2023.10096168

EUSIPCO '22: **Anindya Mondal** *et al.*, "Recovery of Missing Sensor Data by Reconstructing Time-Varying Graph Signals." EUSIPCO 2022. 10.23919/EUSIPCO55093.2022.9909940

ICCVW '21: **Anindya Mondal**, R. Shashant *et al.*, "Moving Object Detection for Event-based Vision using Graph Spectral Clustering." ICCV Workshops 2021. 10.1109/ICCVW54120.2021.00103

Research Experience

Surrey Institute for People-centred AI, CVSSP, University of Surrey — **Guildford, United Kingdom**
Doctoral Researcher — *Oct. 2022 – present*

- Developed a diffusion-based text-to-image model for high-quality exemplar generation in object counting.
- Created a class-agnostic object-counting model using semantic and geometric priors.
- Designed a transformer-based multimodal action-recognition model, improving precision by 50%.
- Led the creation of a benchmark for animal action recognition, detection, and segmentation.

Indian Institute of Science — **Bengaluru, India**
Research Intern — *May 2022 – Aug. 2022*

- Devised a source-free domain-adaptation framework for image classification, enhancing robustness.

Jadavpur University — **Kolkata, India**
Undergraduate Research Assistant — *Oct. 2020 – May 2022*

- Implemented a Sobolev-norm minimisation algorithm for time-varying graph-signal reconstruction.
- Developed a graph-based semi-supervised learning framework for semantic segmentation.
- Improved moving-object detection from event data using graph spectral clustering.

Technical Skills

Languages: Python (expert), MATLAB (expert), C (advanced)

Libraries: PyTorch (expert), scikit-learn (advanced), NumPy (advanced), SciPy (advanced), Pandas (advanced)

Tools: Git (expert), LaTeX (expert), Jupyter Notebook (expert), Docker (advanced)

Awards and Honours

2024: AAAI 2025 Conference & Travel Grant, Philadelphia, USA — conference travel grant.

2023: ICCV 2023 Conference Grant, Paris, France — conference travel grant.

2022: Postgraduate Studentship, University of Surrey, UK — merit-based scholarship for doctoral studies.

2022: Uplink Research Internship Award, ACM SIGKDD India Chapter — internship and travel grant.

Professional Experience

Teaching

2023 & 2024: Teaching Assistant — Applied Machine Learning (M-level), Advanced Topics in Computer Vision & Deep Learning, University of Surrey.

Peer Review

2022–2024: Reviewer — ECCV, NeurIPS, BMVC, IEEE ISBI, ICASSP, IEEE TSP, ICCV Workshops.